Does the Age at Which a Shock Hits Matter?

Violence from before birth to school age, using the end of paramilitary violence in Colombia

Idea 1: statement and assessment

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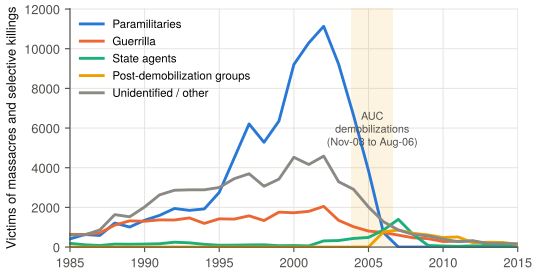
Question

Does the long-run effect of a shock depend on the age at which the child receives it: before birth, in infancy, in preschool, at school age?

- The sensitive-periods view says yes (Cunha and Heckman, 2007, *AER*; Almond, Currie and Duque, 2018, *JEL*).
- The evidence is thick for the prenatal period (famine, pollution, stress) and thin after birth: age profiles are estimated by comparing cohorts hit at different ages by a shock whose timing is rarely exogenous, with people located where they are observed, not where they were hit.
- Violence is the natural shock for this question: large, frequent, well dated. The existing profiles (Burundi, Nigeria, Peru, Germany, Colombia) all use the arrival or intensity of violence and share those two weaknesses.

Setting: the end of violence instead of its arrival



- Paramilitary blocs demobilize one by one, 2003–2006; zero victims after 2007. The end month T_m is observed for each municipality.
- Within a municipality, age at T_m traces the whole profile: born after (never exposed), just before (in utero only), 1–2 years before (infancy), 3–5 (preschool), 6–11 (school age).
- The end date comes from a national negotiation, not from local trends.

Specification

i born in municipality m, cohort c; $k = c - T_m$ is age at the end.

$$Y_{imc} = \sum_{k \neq -1} \beta_k \, \mathbf{1}[c - T_m = k] + \mu_m + \lambda_{c \times d(m)} + \varepsilon_{imc}$$

- β_k is the effect of exposure up to age k, relative to the never-exposed cohort of the same municipality, net of department-by-cohort trends. The profile $\beta_0, \beta_1, \ldots, \beta_{11}$ is the object.
- Staggered timing: Callaway and Sant'Anna (2021); de Chaisemartin and D'Haultfœuille (2024). Validity: pre-conception placebos; Rambachan and Roth (2023) bounds.
- Municipality of birth, individual microdata: GEIH at 20–36 (schooling, employment, earnings); Saber 11 at 17.

What it would add

- **A different source of variation.** Age at the *end* of a shock, with the end set by a national negotiation. It addresses the two weaknesses at once: timing not driven by local trends, location fixed at birth.
- **The whole profile in one design:** prenatal only, infancy, preschool, school age, against never-exposed cohorts of the same municipality.
- **Mechanism:** profile by household displacement (migration module) and by maternal health at birth (vital statistics).

The question is not new. What would be new is a credibly identified postnatal profile.

Assessment

Already well covered. Bundervoet, Verwimp and Akresh (2009, *JHR*); Akresh, Bhalotra, Leone and Osili (2012, *AER P&P*); León (2012, *JHR*); Duque (2024, *JCR*); Akbulut-Yuksel, Tekin and Turan (2025, *JHR*). Same question, same fixed-effects design.

Too little variation. 70% of end dates fall within eighteen months. In-utero-only is a nine-month birth window; never-exposed cohorts become adults in 2026; with current data they exist in 28 municipalities.

Verdict. The only novelty is the instrument, and the instrument is thin. Not a job market paper as it stands.

Appendix. Age at exposure to violence: what the literature finds (1/2)

Paper	Setting	Variation	Outcome	Effect by age
Bundervoet, Verwimp and Akresh (2009, *JHR*)	Burundi, 1993–98	province timing × cohort	height-for-age, children under 5	-0.047 SD per month of exposure
Chamarbagwala and Morán (2011, *JDE*)	Guatemala, 1960–96	department of birth × cohort, three war periods	years of schooling	rural Mayan men: -0.27, -0.71, -1.09 years by period; women: -0.12, -0.47, -1.17; school-age exposure
Shemyakina (2011, *JDE*)	Tajikistan, 1992–98	region × cohort	completion of mandatory schooling	girls of school age 12–13% less likely to complete†
Akresh, Bhalotra, Leone and Osili (2012, *AER P&P*)	Nigeria, Biafra 1967–70	ethnicity × cohort	adult height	losses at every age from birth to adolescence; largest for adolescents
León (2012, *JHR*)	Peru, 1980–93	district of birth × cohort	years of schooling	per year of exposure: in utero -0.07, 0–3 -0.05, 4–6 -0.05, 7+ ≈ 0. Replication with full census (Huaroto, Gonzales and Pacheco, 2026, *JHR*): pre-conception effects significant; only in utero robust

Appendix. Age at exposure to violence: what the literature finds (2/2)

Paper	Setting	Variation	Outcome	Effect by age
Akbulut-Yuksel (2014, *JHR*)	Germany, WWII	city destruction × cohort	schooling, height, earnings	school-age cohorts: about -0.4 years, -1 cm, -6% earnings†
Minoiu and Shemyakina (2014, *JDE*)	Côte d'Ivoire, 2002–07	region × cohort, surveys before, during, after	height-for-age	negative for children in affected regions; channel: household economic losses
Singhal (2019, *JDE*)	Vietnam, 1965–75	bombing intensity, IV	adult mental distress	$+16$ pp severe distress, about 50% of the mean; early-life exposure
Duque (2024, *JCR*)	Colombia, 1980s	municipality-month × cohort, mother FE	dropout, Saber 11	higher homicide rate in early childhood: more dropout, lower scores; 1 SD ≈ -0.1 years†
Akbulut-Yuksel, Tekin and Turan (2025, *JHR*)	Germany, WWII	city bombing × cohort	mental health at 55–70	1 SD bombing: -10% mental-health index, $+16\%$ depression; concentrated under age 5 and in utero

† from a working-paper version. **Reading.** All rows: location × cohort with fixed effects, variation from arrival or intensity. Profiles disagree across settings (worst before school in Peru and Colombia, at school age in Germany and Guatemala, in adolescence for height in Nigeria). Crowded question; unsettled postnatal profile.